

Seasonal SDE using Function OLS

0.1 Empirical Econometric Functions

1. Cao model -

Daily temperature residual follows a k-lag auto-correlation process (Cao)

$$U_t = \sum_{i=1}^k \rho_i U_{t-i} + \sigma \xi_t \quad (1)$$

where

$$\sigma_t = \sigma_0 - \sigma_1 \left| \sin \left(\frac{\pi t}{365} + \varphi \right) \right|$$

and

$$\xi_t \sim \text{i.i.d. } \mathcal{N}(0, 1)$$

2. Campbell model -

Seasonal volatility is approximated by a Fourier series and cyclical volatility by a GARCH model. Temperature model becomes -

$$T_t = \text{Trend}_t + \text{Seasonal}_t + \sum_{l=1}^L \rho_{t-l} T_{t-l} + \sigma_t \varepsilon_t \quad ()$$

where

$$\text{Trend}_t = \sum_{m=0}^M \beta_m t^m \quad \text{and} \quad \text{Seasonal}_t = \sum_{p=1}^P \left(\sigma_{c,p} \cos \left(\frac{2\pi p d(t)}{365} \right) + \sigma_{s,p} \sin \left(\frac{2\pi p d(t)}{365} \right) \right)$$

$$\sigma_t^2 = \sum_{q=1}^Q \left(\gamma_{c,q} \cos \left(\frac{2\pi q d(t)}{365} \right) + \gamma_{s,q} \sin \left(\frac{2\pi q d(t)}{365} \right) \right) + \sum_{r=1}^R \alpha_r (\sigma_{t-r} - \varepsilon_{t-r})^2 + \sum_{s=1}^S \beta_s \sigma_{t-s}^2 \quad (2.11)$$

where $\varepsilon_t \sim \text{i.i.d.}(0, 1)$ and $d(t)$ is a repeating function that cycles 1 to 365.

3. Alaton Model -

Mean reverting process with monthly varying volatility -

$$dT_t = \frac{dT_m}{dt} + a(T_m^t - T_t)dt + \sigma dW_t \quad (2)$$

where

$$T_m^t = A + Bt + C \sin(wt + \varphi)$$

and $w = \frac{2\pi}{365}$, and φ is the phase angle.

4. Benth Model -

CAR(p) model, which is a continuous AR(p) time series representation, where temperature can be modeled as -

$$T(t) = \Lambda(t) + X_1(t) \quad (3)$$

where $\Lambda(t)$ is a deterministic seasonal function representing average temperature and the stochasticity is modeled through $X(t)$, which is a mean reverting Ornstein-Uhlenbeck process $\in \mathbb{R}^p$ for $p \geq 1$:

$$dX(t) = AX(t)dt + e_p\sigma(t)dB(t) \quad (4)$$

where $B(t)$ is a Brownian motion, e_k is the k -th unit vector in $\mathbb{R}^p$, $k = 1, \dots, p$, $\sigma_t > 0$ is a real-valued and square-integrable function and A is a $p \times p$ matrix of the form:

$$A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -\alpha_p & -\alpha_{p-1} & -\alpha_{p-2} & \cdots & -\alpha_1 \end{bmatrix}$$

with explicit solution for $X(t) = x \in \mathbb{R}^p$ -

$$X(S) = \exp(A(S-t))x + \int_t^S \exp(A(S-u))e_p\sigma(u)dB(u) \quad (5)$$

$s \geq t$

5. Cabrera Model -

Average temperature t -th day:

$$T_t = \frac{T_{\max_t} + T_{\min_t}}{2}$$

following an Ornstein-Uhlenbeck (OU) process:

$$dT_t = a(T_m - T_t)dt + \sigma_t dW_t$$

Seasonality with a time-dependent mean that accounts for this seasonality implies -

$$\frac{dT_t^m}{dt} = B + \omega \cdot C \cdot \cos(\omega t + \phi)$$

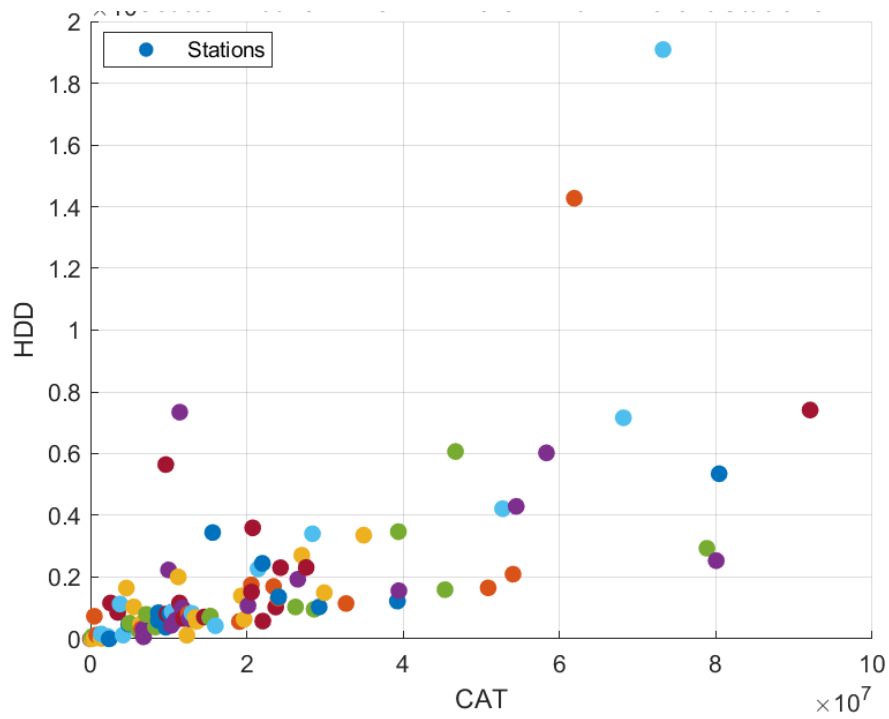


Figure 1: Scatter plot - Temperature indices comparison (CAT vs HDD) across all weather stations in Hong Kong over the period 2010 - 2024

0.2 SDE -

$$dT_t = \left\{ \frac{dT_t^m}{dt} + a(T_t^m - T_t) \right\} dt + \sigma_t dW_t$$

Distribution of differences $\Delta T_t = T_t - T_{t-1}$ has mean and variance :

$$\overline{\Delta T_t} = \frac{1}{n} \sum_{t=1}^n \Delta T_t$$

$$\frac{1}{n} \sum_{t=1}^n (\Delta T_t - \overline{\Delta T_t})^2.$$

Calibration

Following Alaton as well as Bibby and Sorensen [?] [?] in a *martingale estimation method* for drift and diffusion coefficients. This efficient estimator can be obtained by finding the value of $\hat{a}_n$ such that:

$$G_n(a) = 0 = \sum_{i=1}^n \frac{\partial_a b(T_{i-1}; a)}{\sigma_{i-1}^2} (T_i - \mathbb{E}[T_i | T_{i-1}])$$

Discretized SDE based variance estimator -

$$\hat{\sigma}_u^2 = \frac{1}{N_u - 2} \sum_{j=0}^{N_u-1} (T_j - \hat{a}T_{j-1}^m - (1 - \hat{a})T_{j-1})^2$$

0.3 OLS to be fitted -

$$T_t^m = a_1 + a_2 t + a_3 \sin(\omega t) + a_4 \cos(\omega t)$$

where:

- T_t^m : Modeled temperature at time t
- ω : Angular frequency, typically set to $\frac{2\pi}{365}$ for daily data with an annual cycle

such that -

$$T_t^m = A + Bt + C \sin(\omega t + \phi)$$

- $A = a_1$, $B = a_2$, $C = \sqrt{a_3^2 + a_4^2}$
- $\phi = \arctan\left(\frac{a_4}{a_3}\right) - \pi$

0.4 Alternate - Jump Model

0.4.1 Parameter Calibration

Using rolling window estimation, the simulation was designed to be run in two dimensions. Spatial one - run on different locations. Temporal one - to capture changes in the parameters through time for each location. Simulations were run using 5 years initial period and then following iterations added one more year to previous one. In each turn error estimates, HDD and CDD estimates were calculated. Results were compared with actual HDD and CDD values. A turn, on the other hand, consisted of 10000 runs. At the end of each run, average values of 10000 runs were calculated to estimate the parameters. with whole process repeated using initial 6,7,8,... years of data. Finally, parameters of the year that offered best estimates of the HDDs and CDDs were chosen for predictions.

Discretization Discretization was done by using Euler approximation as following:

$$T_{t+1} = T_t + T_{t+1}^m + T_t^m + b(T_t^m - T_t) + H_{t+1} + Y_t(1-\alpha) + (Q_{t+1} - Q_t) + Z_t(1-\beta) + (R_{t+1} - R_t) \quad (6)$$

where -

- $H_{t+1} = \sqrt{\gamma_0 + \gamma_1 H_t \epsilon_t}$, γ_0 and γ_1 are ARCH parameters
- $\epsilon \sim N(0, 1)$
- T_t^m is estimated as per previous OLS estimation (0.3)

Mean reversion Mean reversion parameter b were estimated as:

$$\hat{b} = -\log \left(\frac{\sum_{i=1}^n \frac{(T_{i-1} - T_{i-1}^m)(T_i - T_i^m)}{\sigma_{i-1}^2}}{\sum_{i=1}^n \frac{(T_{i-1} - T_{i-1}^m)^2}{\sigma_{i-1}^2}} \right)$$

Jump component

Mean reversion parameters ([?]) of jump parts were estimated as $\hat{\alpha} = 1$ and

$$\hat{\beta} = -\log \left(\frac{\sum_{i=1}^n x_{i-1} x_i}{\sum_{i=1}^n |X_{i-1}|^2} \right)$$

Jump intensity

$\hat{\lambda} = j/N$ where j is the number of jumps and N is the total number of observations.

Standalone simulations of jumps were added as shocks to the discretized temperature simulation.

1 Results

Jump Model Parameters -

- Linear regression coefficients (A, B, C , phase)
- GARCH parameters (α_0, α_1)
- Jump process parameters (fast and slow jump intensities, means, and standard deviations)
- Mean reversion parameter (β)
- Heating Degree Days (HDD) and Cooling Degree Days (CDD) metrics
- Mean reversion coefficient (b)

Parameter	Description
A	Intercept of the linear regression model. Represents the baseline temperature when time is zero.
B	Slope coefficient of the linear trend. Indicates the rate of temperature change over time.
C	Amplitude of the sine component capturing daily seasonal variations.
Phase	Phase shift of the seasonal component. Adjusts the position of the sine wave to align with actual seasonal peaks and troughs.
α_0	GARCH parameter representing the constant term in the conditional variance equation.
α_1	GARCH parameter representing the effect of the previous period's squared residual on current volatility.
Fast_Intensity	Intensity (frequency) of fast jumps in temperature data.
Mean_Fast	Mean size of fast jumps.
SD_Fast	Standard deviation of fast jumps.
Slow_Intensity	Intensity (frequency) of slow jumps in temperature data.
Mean_Slow	Mean size of slow jumps.
SD_Slow	Standard deviation of slow jumps.
β	Mean reversion parameter. Measures the speed at which temperature deviations revert to the mean.
HDD	Accumulated Heating Degree Days over the window. Calculated as the sum of $(65 - \text{Temperature})$ for observations with Temperature $\geq 65^\circ\text{F}$.
HDD_exact	Exact Heating Degree Days calculated directly from the temperature data.
CDD	Accumulated Cooling Degree Days over the window. Calculated as the sum of $(\text{Temperature} - 65)$ for observations with Temperature $\geq 65^\circ\text{F}$.
CDD_exact	Exact Cooling Degree Days calculated directly from the temperature data.
b	Mean reversion coefficient derived from the mean reversion function.
Difference	Metric reflecting the sum of squared differences between simulated and actual temperatures.

1.1 Sample Results

Model Calibration results using Temperature observations from Station 6016 - German Swiss International School district. Data availability determines number of rows corresponding to a specific time window iteration.

Table 1: Sample Results from Model Fitting - Iteration 1

Parameter	Value
Iteration	1
A	15.64477372
B	0.001316788779
C	0.231485769
Phase	-3.681740454
Arch_Param1	1.30397959
Arch_Param2	0.9983245385
Jump_Param1	0.0004759638267
Jump_Param2	0.156545852
Jump_Param3	0.09574271078
Jump_Param4	0.003212755831
Jump_Param5	-1.384458488
Jump_Param6	1.602955686
β	0.02563029293
HDD	481562.8236
HDD_exact	368274.9
CDD	0
CDD_exact	0
b	0.01264089367
Difference	1701779.746

2 References

R code ref link

P. Alaton, B. Djehiche, and D. Stillberger, On modelling and pricing weather derivatives, *Applied Mathematical Finance*, 9(1), pp. 1–20, 2002.

B. M. Bibby and M. Sørensen, Martingale estimation functions for discretely observed diffusion processes, *Bernoulli*, pp. 17–39, 1995 3. S. M. Iacus, *Option Pricing and Estimation of Financial Models with R*, John Wiley Sons Ltd, 2011